USER MANUAL

BYB TECH

CHRONO

bybtech.it

INDEX

Quick start	Pag. 3
What's in the box?	Pag. 4
Device	Pag. 5
Chrono: mounting options	Pag. 7
Chrono: device usage	Pag. 8
Chrono: device configuration	Pag. 9
Lap type definition	Pag. 14
External trigger: track positioning	Pag. 15
BLE sensors configuration	Pag. 18
External trigger: device configuration	Pag. 20
Wireless connection	Pag. 22
Battery charging	Pag. 24
How to wash	Pag. 25
Troubleshooting	Pag. 26

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QUICK START

Quick start

- Mount BYB Chrono:
  - Use the included plastic mount and Velcro strap
  - Place it securely on your frame or helmet
  - Make sure the top of the device faces the sky

- Power On and wait for the GPS fix:
  - Press the Power button once (power LED solid green)
  - Wait for the Run LED to turn solid green (GPS fix acquired - First fix can require up to 5 minutes)

- Start recording:
  - Press the Run button once
  - Run LED blinks slowly when recording
  - Press again to stop recording

- Connect to the App:
  - Press and hold Run Button for 2 seconds to enable Wi-Fi (Run LED blinking fast)
  - Connect to Wi-Fi network: BYB-DeviceName (default password: bybchrono).
  - Open the BYB app > Device Page to manage settings and download your runs.

- Optional setup:
  - Add external trigger or BLE sensors via app.
  - Configure lap type, wheel size, and other advanced features.

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WHAT'S IN THE BOX?

Components list

BYB Chrono is a complete lap timing system that is composed by:

- x1 BYB Chrono device
- x1 Plastic mounting
- x3 M3x5 screws
- x1 Velcro strap
- x1 Speed sensor magnet
- BYB Chrono external triggers (optional)

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DEVICE

Main features

BYB Chrono is an accurate lap timer device with the following features:

- High accuracy GPS antenna
- Fast GPS update rate (25Hz)
- Multi-band and multi-constellation GPS module
- Built-in speed sensor
- Compatible with BYB Chrono external triggers
- BLE heart rate and power meter sensors compatibility
- Compact and lightweight
- Waterproof (IP65)
- Wireless connection
- Smartphone app (iOS and Android)

BYB Chrono can be used in different configurations:

- GPS
- GPS + Speed sensor
- GPS + External trigger
- GPS + Speed Sensor + External trigger

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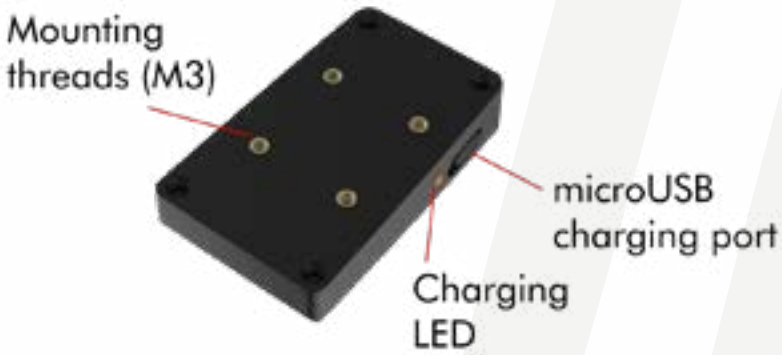
DEVICE

Main features



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DEVICE

Mounting options

BYB Chrono can be installed anywhere on your bike, motorcycle, or vehicle. For optimal performance, it is recommended to position the top of the device—or the side with the button interface—facing upward toward the sky.

BYB Chrono comes with a plastic mount and a Velcro strap for easy installation. Below are some examples of possible mounting options:

[CAUTION]
For extra safety, it is recommended to secure the device with an additional piece of tape to prevent it from coming loose in case of crashes or strong impacts.

[CAUTION]
Always check the setup/mounting stability before every ride.

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DEVICE USAGE

Chrono buttons and functionalities

Switch the device on/off
- To switch on: Press the power button. The power LED will blink once, and the device will be ready.
- To switch off: Press and hold the power button for 1.5 seconds.

Starting/stopping an acquisition
- To start acquisition: Press the run button once. The run LED will start blinking slowly.
- To stop acquisition: Press the run button again. The run LED will stop blinking.

Connection mode
- Press and hold the run button for 2 seconds to enable Wi-Fi connection mode.
- Press the button again to disable the connection.

Resetting the system
- Factory reset: Press and hold the power button for 8 seconds to restore default settings.
- Force restart: Press and hold the power button for 15 seconds. The device will restart automatically.

Power LED function
- On: The device is powered on.
- Blinking: Battery level is low (below 40%).

Run LED function
- Off: The device is ready to acquire, but GPS does not have a fix.
- On: The device is ready to acquire, and GPS has a fix.
- Blinking slowly: The device is sampling data.
- Blinking quickly: The device is in connection mode.

[CAUTION]

A GPS fix is always required for proper operation. Make sure to place the device with the top facing the sky and wait until the Run LED turns on before starting any acquisition.

[TIPS]

While waiting for a GPS fix, keep in mind that the device will automatically power off after 10 minutes of inactivity. It is recommended to keep the device powered on and facing the sky to acquire as many satellites as possible and improve overall accuracy.

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DEVICE CONFIGURATION

Chrono configurations

Device name
This is the name assigned to the device. When connection mode is active, this name will appear as the Wi-Fi network name (e.g., BYB-DeviceName).

Rider name
The name of the rider associated with the device.

Lap type
Select between a closed circuit (motocross or xc loop) and an open circuit (downhill track).

External trigger
Enable or disable the use of an external trigger.

External trigger ID
The ID assigned to the external trigger(s). It must match the ID of the external trigger(s) being used.

Speed sensor
Enable or disable the use of a speed sensor.

Wheel diameter
Enter the wheel size, used to calculate speed and distance based on sensor data.

Heart rate sensor
Enable or disable the heart rate sensor. Enabling this option will activate Bluetooth Low Energy.

Power meter sensor
Enable or disable the power meter sensor. Enabling this option will activate Bluetooth Low Energy.

IMU orientation
Set the orientation of the device. This is used for calculating advanced metrics.

Device password
This is the password for accessing the device's Wi-Fi network. The default password is "bybchrono".

Format SD
This command will wipe all the runs stored inside the memory.

Factory reset
This command will reset the entire device to the factory configurations.

DEVICE CONFIGURATION

ONLY GPS

The basic configuration of the device is defined by the GPS working mode.
To properly load route data into the app, the GPS must always be active. It is crucial to wait for a GPS fix before using the device. The longer the device remains powered on, the more satellites it can connect to—significantly improving the accuracy of the GPS signal.

The GPS mode allows the device to record both the full route and speed data. To enable only the GPS mode, make sure to:
- Disable the speed sensor (select NO)
- Disable the external trigger (select NO)

[CAUTION]
A GPS fix is always required. Always wait for the Run LED to turn on before starting a run.

[TIPS]
For a quicker and more accurate GPS fix, place the BYB Chrono with the top facing the sky. The first GPS fix may take a few minutes. However, once acquired, subsequent hot-fixes (within 2–3 hours of use) will take just a few seconds.

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DEVICE CONFIGURATION

GPS + SPEED SENSOR

This advanced configuration leverages the speed sensor to provide accurate speed and distance calculations. This mode is especially useful when the track is in wooded areas, where GPS reception may be weak. By ensuring precise distance measurement, it allows for reliable comparisons between runs.

The GPS + speed sensor mode captures both the full route and the speed sensor data.
To enable this mode:
- Enable the speed sensor (select YES).
- Disable the external trigger (select NO).

[TIPS]
If using the speed sensor, the magnet can be installed directly on the spokes using a couple of thin cable ties. The two magnets can be stacked togheter in order to increase the detection range.

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DEVICE CONFIGURATION

GPS + EXTERNAL TRIGGER

This advanced configuration utilizes external triggers to provide accurate timing. The triggers can be switched on and placed along the track to record the exact time when they are crossed.

For open laps, it is mandatory to place at least two triggers: one at the start and one at the finish. For closed laps, only one trigger is required at the start/finish line. Additional triggers can be added as intermediate splits. The total number of triggers used must be specified in the app.

The GPS + Speed Sensor Mode captures both the full route and accurate timing data.
To enable this mode, connect to your BYB Chrono device and navigate through the "settings" page:

- Enable the speed sensor (select NO).
- Disable the external trigger (select YES).

[CAUTION]
The external trigger mode reduces the battery life of the BYB Chrono. It should only be enabled

[TIPS]
For an open lap, avoid starting and stopping too close to the external triggers. For a closed lap, position the trigger as far as possible from intersections.

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DEVICE CONFIGURATION

GPS + SPEED SENSOR + EXTERNAL TRIGGER

This advanced configuration combines both the speed sensor and external trigger to provide accurate speed, distance, and timing data. It is the most precise and comprehensive mode, offering valuable insights for both timing and comparisons.

For open laps, it is mandatory to place at least two triggers: one at the start and one at the finish. For closed laps, only one trigger is required at the start/finish line. Additional triggers can be added as intermediate splits. The total number of triggers used must be specified in the app.

To enable this mode, connect to your BYB Chrono device and navigate through the "settings" page:
- Enable the speed sensor (select YES).
- Enable the external trigger (select YES).

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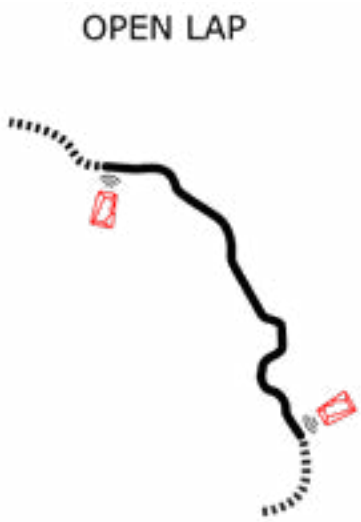
DEVICE CONFIGURATION

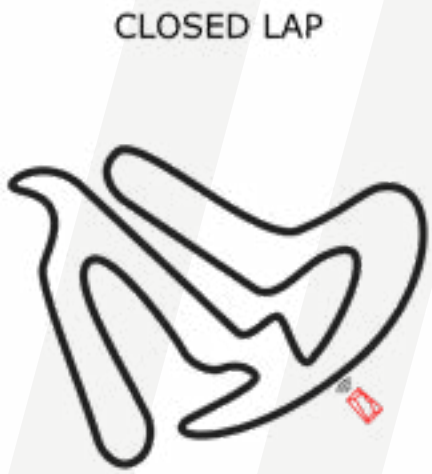
LAP TYPE

BYB Chrono supports two types of laps: open lap and closed lap.

Open lap: The track has a distinct start and finish, with the entry and exit points being separate.
Closed lap: The track forms a loop with a shared start/finish line. In this mode, both the entry and exit laps are automatically excluded from any calculations.

Additional triggers can be placed as intermediate splits, and the total number of triggers used must be specified in the app.





[TIPS]
The external trigger is optional. The GPS alone is sufficient to define custom start and finish waypoints via the app, making it adaptable to various track configurations.

[CAUTION]
Before each run, always specify the type of lap being used.

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DEVICE CONFIGURATION

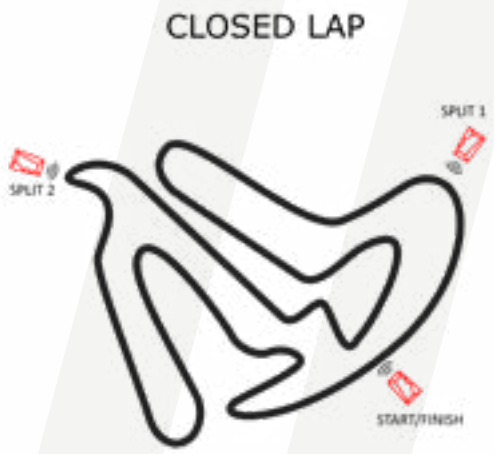
EXTERNAL TRIGGER POSITIONING

To ensure the best performances, it's crucial to position the external triggers properly.

Open lap: Place the triggers approximately 50 meters after the starting point and 50 meters before the stopping point. All additional intermediate triggers must be placed at a proper distance from each other to avoid interference.

Closed lap: The track forms a loop with a shared start/finish line. In this mode, both the entry and exit laps are automatically excluded from any calculations. All additional intermediate triggers must be placed at a proper distance from each other to avoid interference (more than 100 meters).

A good triggers placing example can be the following one:



[TIPS]

Additional triggers can be placed as intermediate splits, and the total number of triggers used must be specified in the app.

[CAUTION]

It is crucial to position each trigger as far apart as possible to avoid interference.

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DEVICE CONFIGURATION

EXTERNAL TRIGGER ID

BYB Chrono can be paired with an external trigger to achieve highly accurate timing, independent of GPS reception quality.

The trigger's name can be customized. A unique name must be entered (default: "Chrono") that matches the trigger's ID.

[CAUTION]

The external trigger ID set in the BYB Chrono configuration must match the ID configured in the external trigger itself. If more triggers are used, all the external triggers must share the same ID.

[TIPS]

The triggers can be shared between more Chrono devices.

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DEVICE CONFIGURATION

WHEEL SIZE

BYB Chrono supports different wheel sizes to accommodate all possible diameters.

You can either select a standard wheel size from the list or enter a custom diameter (in millimeters) for precise speed and distance calculations. The wheel size setting refers to the wheel used by the built-in speed sensor to measure speed.

When the speed sensor is enabled, it is mandatory to set the Wheel Size field to YES.

[TIPS]

Place the magnet on the disc or on the spokes as much closer as possible to the center of the hub.

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DEVICE CONFIGURATION

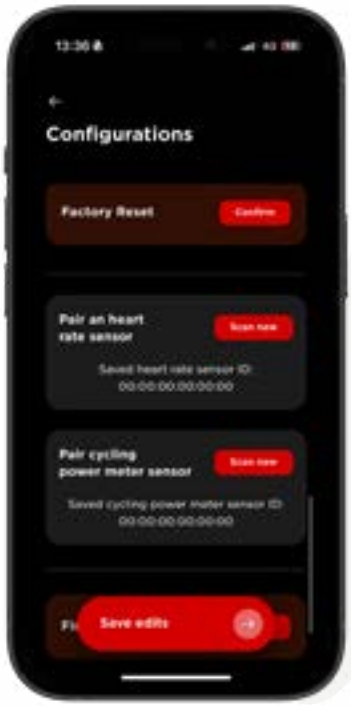
BLE SENSORS (HEART RATE and POWER METER)

BYB Chrono can connect to commercial BLE heart rate monitors and cycling power meters.

Before use, each sensor must be paired with BYB Chrono. Once paired, the sensor's unique ID will be saved in the device for future sessions.

To pair a heart rate or power meter sensor:
- Activate the sensor: heart rate sensor (by wearing it) or the power sensor (by moving it).
- Enable WiFi mode on the BYB Chrono.
- Open the app and navigate to the device configuration menu.
- Tap "Scan heart rate sensor" or "Scan cycling power meter sensor".
- Wait for the pairing process to complete — the unique code of the paired sensor will be displayed.

Before starting a run, make sure to set "Heart rate sensor" or "Power meter sensor" to YES in the device configuration menu.



[CAUTION]
Using a BLE sensor will activate the BLE, therefore it will reduce the battery life.

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DEVICE CONFIGURATION

IMU ORIENTATION

BYB Chrono provides the possibility of selecting a specific IMU orientation.
This feature is used to select how BYB Chrono is installed. These configurations are used to know the direction of each vector (X, Y, Z) and perform advanced calculations.

[CAUTION]
The directions to be defined, according to the image above, are: X, Y, Z.

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DEVICE USAGE

Trigger buttons and functionalities

Switching the device on/off
- To switch on: Press the power button. The power LED will start blinking slowly.
- To switch off: Press again the power button.

Resetting the system
- Factory reset: Press and hold the power button for 8 seconds to restore default settings.
- Force restart: Press and hold the power button for 15 seconds. The device will restart automatically.

Power LED function
- Blinking LED: Indicates that the trigger is active.

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DEVICE CONFIGURATION

Trigger configurations

The external trigger has several configurations:

External trigger ID
This is the unique identifier assigned to the device. The WiFi network will appear as “TRIGGER-EXTERNAL-TRIGGERI-D”. Make sure this ID is also correctly entered in the BYB Chrono configuration to enable proper communication between the devices.

Device password
This is the password for accessing the device’s Wi-Fi network. The default password is “bybchrono”.

Factory reset
This command will reset the entire device to the factory configurations.

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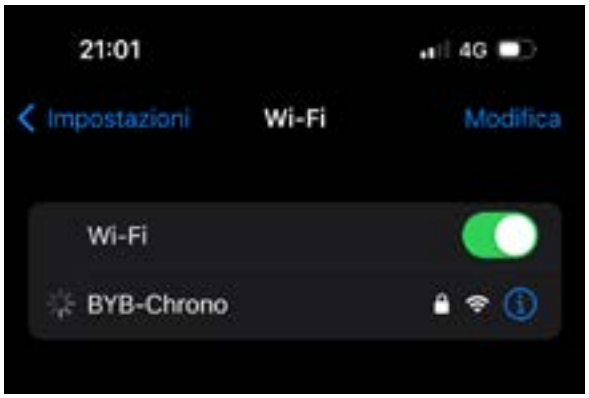
WIRELESS CONNECTION

Enable the wireless connection on Chrono

- Switch on the device and press for 2 seconds the run button. The run LED starts blinking fast.
- Search for the Wi-Fi network called "BYB-Chrono" (if default).
- Login with the password "bybchrono" (if default).
- The connection menu may display "Connected" or "Connection without internet"
- Once connected, open the app into the "device page" and wait for the loading of the data. Now it's possible to manage the configurations and download the acquired runs.



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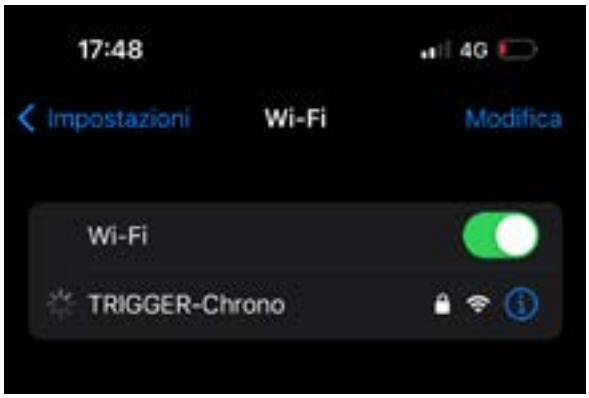
WIRELESS CONNECTION

Enable the wireless connection on the external trigger

- Switch on the device. The power LED start blinking.
- Search for the Wi-Fi network called “TRIGGER-Chrono” (if default).
- Login with the password “bybchrono” (if default).
- The connection menu may display “Connected” or “Connection without internet”
- Once connected, open the app into the “device page” and wait for the loading of the data. Now it’s possible to manage the configurations.



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BATTERY CHARGING

Charge the device

Both Chrono and the external trigger have a small capacity battery. To recharge them:
- Remove the micro USB protective cap.
- Connect the micro USB cable to the micro USB port of the device.
- Use ONLY an USB PC port or a smartphone charger with an output of 5V, 1A.
- The charging LED is on when recharging the battery.
- The charging LED is off when the battery is fully charged.
- Insert the microUSB protective cap before the next ride

[CAUTION]
Don't connect the microUSB port to an input greater than 5V. Use only phone chargers or PC USB port.

[TIPS]
To increase battery life recharge the system only before every use.

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HOW TO WASH THE DEVICE

Device cleaning

BYB Chrono is an electronic device, therefore it must be clean without any pressure washer. Clean it gently with water and a soft cloth.

[CAUTION]
Don't use pressure washer directly on the device.

[CAUTION]
Check if the micro USB connector is dry after cleaning the device and before charging it again.

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TROUBLESHOOTING

The WiFi password is wrong

If you're unable to connect to the BYB-Chrono Wi-Fi network due to an invalid password, try:
- Try logging in with the default password: "bybchrono"
- If you previously changed the password and can't remember it: factory reset the device (press and hold the power button for 8 seconds), once reset, connect again using the default password: bybchrono

The WiFi connection is unstable

Some smartphones may automatically disconnect from the BYB-Chrono Wi-Fi network because it doesn't provide internet access. To prevent this, the best solution is to set the BYB-Chrono Wi-Fi network as a static connection in your phone's Wi-Fi settings. This ensures a stable connection during configuration and data transfer. Set:
- IP address as static: 192.168.1.1
- Subnet mask: 255.255.0.0 (if asked)

Chrono can't get a GPS fix

To ensure that the BYB Chrono acquires a proper GPS fix, we recommend switching on the device a few minutes before starting your run. Leave the Chrono powered on for 2 to 10 minutes, placed in a steady position with a clear view of the sky.
Once the device has acquired a reliable 3D GPS fix, the Run LED will turn solid green.

Remember: The longer Chrono searches for satellites, the more satellites it connects to—resulting in more accurate GPS positioning.

Can't see the GPS map on the acquired data

If the GPS map is missing from your run, it likely means that data logging started before a solid GPS fix was acquired. Always wait for the Run LED to turn solid green before starting an acquisition.

Chrono is not responsive

If the device becomes unresponsive, press and hold the power button until it reboots (approximately 15 seconds).

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